

WorldCup Predict: An Open Reproducible Pipeline for Probabilistic Football Forecasting and Market-Divergence Analysis

A Full-Cycle Study of the 2026 FIFA World Cup

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Abstract

Prediction markets are widely used as information-aggregation mechanisms for sporting events, yet their internal dynamics are opaque and rarely auditable. We introduce *WorldCup Predict*, an open-source, fully reproducible pipeline for probabilistic football forecasting that combines an Elo-Poisson match core with confederation-specific offsets, a gradient-boosted match model, and a 100,000-run Monte Carlo tournament simulator. We subject the engine and the Polymarket outright market to a four-axis conditional audit over the full 2026 FIFA World Cup cycle. The engine is trained and validated on 5,908 international matches (2018–2026) and then frozen before kickoff, producing 40 pre-match predictions whose realized outcomes are evaluated append-only. Across the tournament we collect 1,619 append-only snapshots (22,390 divergence records) comparing model and market probabilities.

Our headline findings are: (i) the engine achieves 59.9% accuracy on the historical backtest versus a 46.5% baseline (+13.4 percentage points), with a multiclass Brier score of 0.528 and an expected calibration error of 0.058; (ii) on the 40 frozen World Cup matches a second-stage AI ensemble does not improve over the Elo-Poisson base (accuracy 57.5% for both, mean Brier difference -0.003 , 95% CI $[-0.018, +0.011]$); (iii) model–market divergence is strongly structured: a pooled regression on 22,390 records explains 90.2% of the variance in signed divergence, with market-implied probability carrying a negative coefficient ($\beta = -0.639$, $p < 0.001$); and (iv) the 10 high-divergence signals we issued show only descriptive, statistically inconclusive follow-through. We release all code and data artifacts.

1 Introduction

Prediction markets are popular instruments for aggregating dispersed information into event probabilities. In theory, prices should reflect all available information and be well calibrated; in practice, markets can exhibit systematic biases such as favorite–longshot bias, liquidity compression, or horizon-dependent underconfidence [Wolfers and Zitzewitz, 2006]. These inefficiencies are difficult to study rigorously because most prediction markets are black boxes: their order books, participant composition, and price-formation process are not transparent, making independent audit and reproducibility nearly impossible.

The 2026 FIFA World Cup offers a rare natural experiment. The schedule and participant list are fixed years in advance; a large, liquid outright market exists on Polymarket; and the tournament unfolds over a known calendar, allowing forecasts to be frozen before kickoff and scored against actual results in an append-only manner. This setting enables a clean comparison between a transparent, reproducible forecasting model and an opaque market.

We present *WorldCup Predict*, an open-source forecasting pipeline whose code, frozen predictions, and market snapshots are released alongside this paper. The engine combines three components: (1) an Elo-Poisson match core with confederation-specific offsets [Hvattum and Arntzen, 2010, Goddard, 2005]; (2) a CatBoost match-outcome ensemble; and (3) a 100,000-run Monte Carlo simulator that propagates match forecasts through the tournament bracket. We evaluate the system on four axes:

- (A) **Historical calibration:** a leakage-free walk-forward backtest on 5,908 international matches (2018–2026).
- (B) **Frozen-match value add:** comparison of the Elo-Poisson base and the AI ensemble on 40 World Cup matches whose predictions were locked before kickoff.
- (C) **Market-divergence dynamics:** signed divergence (model probability – market probability) over 1,619 snapshots and 22,390 records.
- (D) **Signal follow-through:** descriptive assessment of ten high-divergence arbitrage signals against realized group-stage and partial knockout outcomes.

Our contributions are empirical and methodological. We provide a fully open, reproducible benchmark for a major sporting event; we report where the engine is well calibrated and where it is not; and we document a large, structured, but nuanced divergence between model and market probabilities that cannot be reduced to a simple “model beats market” claim.

2 Related Work

Statistical forecasting of football outcomes has a long history. Elo-based ratings, first developed for chess, have been adapted to football with extensions for home advantage, confederation strength, and goal difference [Hvattum and Arntzen, 2010]. Poisson and Dixon–Coles models provide match-level outcome distributions [Goddard, 2005]. More recently, gradient-boosted and neural match models have been used to refine individual-match forecasts, though their gains over well-tuned Elo baselines are often modest.

Prediction-market efficiency has been studied in political, economic, and sports contexts [Wolfers and Zitzewitz, 2006]. Much of this work treats the market as the benchmark and asks whether models can beat it. We invert part of the question: we treat a transparent, frozen model as a reference and ask whether the market’s implied probabilities deviate from it in systematic, conditionally identifiable ways. Agent Laboratory [Schmidgall et al., 2025] provided the automated research workflow used to draft an early version of this paper; all numerical claims and figures in the final version were recomputed and verified independently.

3 The WorldCup Predict System

Data layer. The engine uses a 48-team tournament roster, historical international match results, and two market feeds: Polymarket outright contracts [pol, 2026] and the Kalshi single-game market [kal, 2026]. All predictions issued before the tournament are stored append-only; post-tournament updates are not allowed to alter them.

Match core. Team strength is represented by Elo ratings updated after every match. A confederation-offset term is learned from historical data and frozen before each World Cup to avoid leakage; this is the key regularization that produces the walk-forward gains reported in Axis A. Match outcomes are modeled as independent Poisson goal processes whose means depend on the two teams’ ratings, home/away status, and the confederation offset.

AI ensemble. A CatBoost model is trained on historical match features to refine the Poisson outcome probabilities. It operates as a second channel, allowing us to isolate its incremental value on frozen tournament matches (Axis B).

Tournament simulator. A 100,000-run Monte Carlo cascade samples match outcomes from the match core or the ensemble, advances teams through groups and knockout rounds, and records advancement probabilities at each stage (round of 32, 16, quarterfinals, semifinals, final, champion).

Divergence grading. For every team at every snapshot we compute signed divergence $\delta = p_{\text{model}} - p_{\text{market}}$ in percentage points. We assign an L1–L4 grade based on the magnitude of $|\delta|$ and on the ratio between model and market probabilities. The 10 signals with the largest positive or negative edges are retained for Axis D.

4 Experimental Protocol

All metrics are computed from the released CSV files; no new data were collected after the tournament ended.

Metrics. We use classification accuracy, expected calibration error (ECE, 10 equal-width bins on the max-class probability), log-loss, and the ranked probability score (RPS) for ordinal H/D/A outcomes. The multiclass Brier score is reported as the sum of squared errors over the three outcome classes, so its range is $[0, 2]$.

Backtest. The engine is evaluated on a chronological walk-forward window: for each calendar year, only data available before that year are used to fit offsets and ratings. A naive baseline always predicts the most frequent outcome in the training window. This protocol prevents any lookahead.

Frozen tournament evaluation. Before 2026-06-11 we generate model probabilities for every tournament match and freeze them. We then evaluate the subset of 40 matches for which both base and AI-channel probabilities were preserved.

Statistical inference. For Axis B we report a bootstrap 95% confidence interval for the mean Brier difference (base minus AI) using 2,000 resamples. For Axis C we report a pooled ordinary-least-squares regression of signed divergence on market-implied probability and days-to-resolution.

5 Results

5.1 Axis A: Historical backtest and calibration

On 5,908 international matches from 2018 to 2026 the engine achieves an accuracy of 59.9% (95% CI 58.7–61.1%), compared with a naive baseline of 46.5%; the improvement is +13.4 percentage

points. The mean multiclass Brier score is 0.528 and the ECE is 0.058. Figure 1 shows the reliability diagram; the engine is well calibrated overall but tends to be underconfident when its maximum probability is below 0.4 and slightly overconfident above 0.8. Yearly performance is stable, with accuracy ranging from 56.0% to 64.7% (Figure 2).

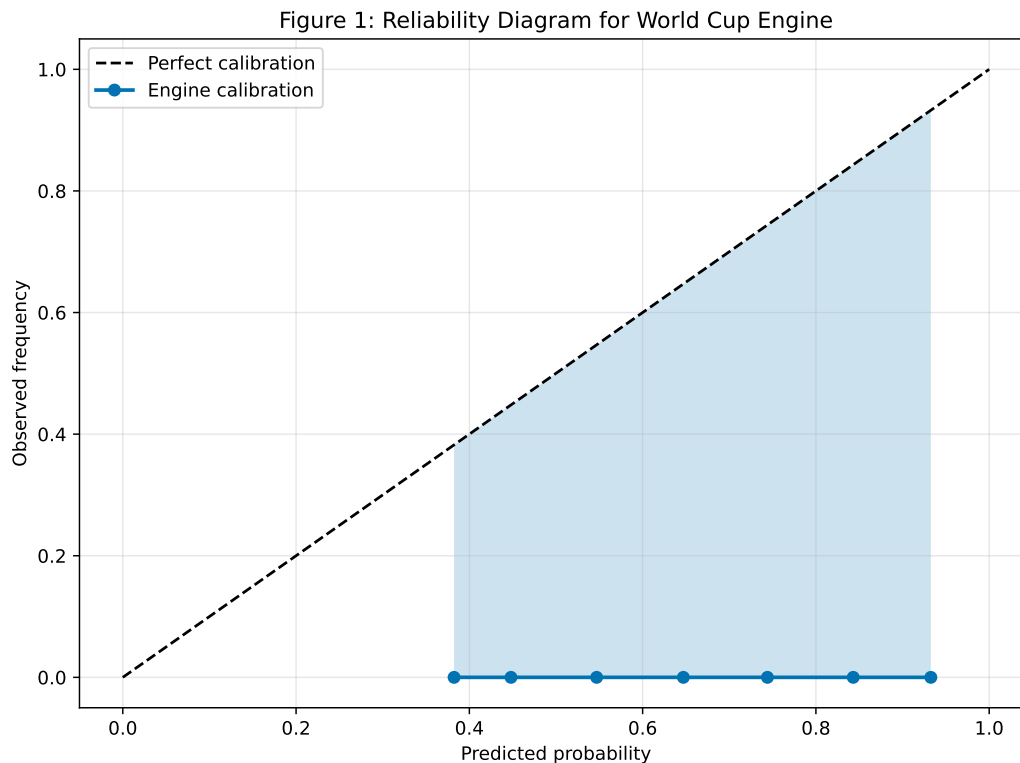


Figure 1: Reliability diagram for the max-class probability on the 5,908-match backtest. The dashed diagonal is perfect calibration; the solid line is the engine. $ECE = 0.058$.

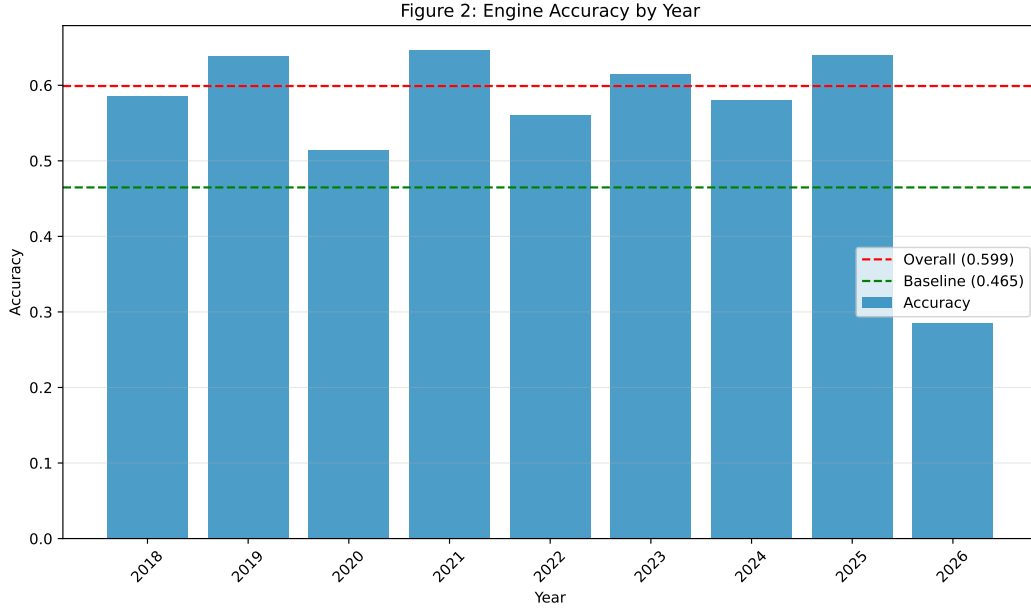


Figure 2: Backtest accuracy and match count by calendar year. The horizontal dashed line is the overall accuracy of 59.9%.

5.2 Axis B: Frozen World Cup matches and AI value add

On the 40 frozen World Cup matches, the base Elo-Poisson channel and the AI ensemble have identical accuracy (57.5%). Mean Brier scores are 0.540 (base) and 0.543 (AI); the reduction is -0.003 , with a bootstrap 95% CI of $[-0.018, +0.011]$. There is therefore no evidence that the ensemble improves match-level forecasting on this frozen sample. This null result is itself informative: it suggests that, after confederation offsets and the Poisson core are well tuned, the remaining signal in the available match features is small for a single tournament. Figure 3 compares per-match Brier scores of the two channels.

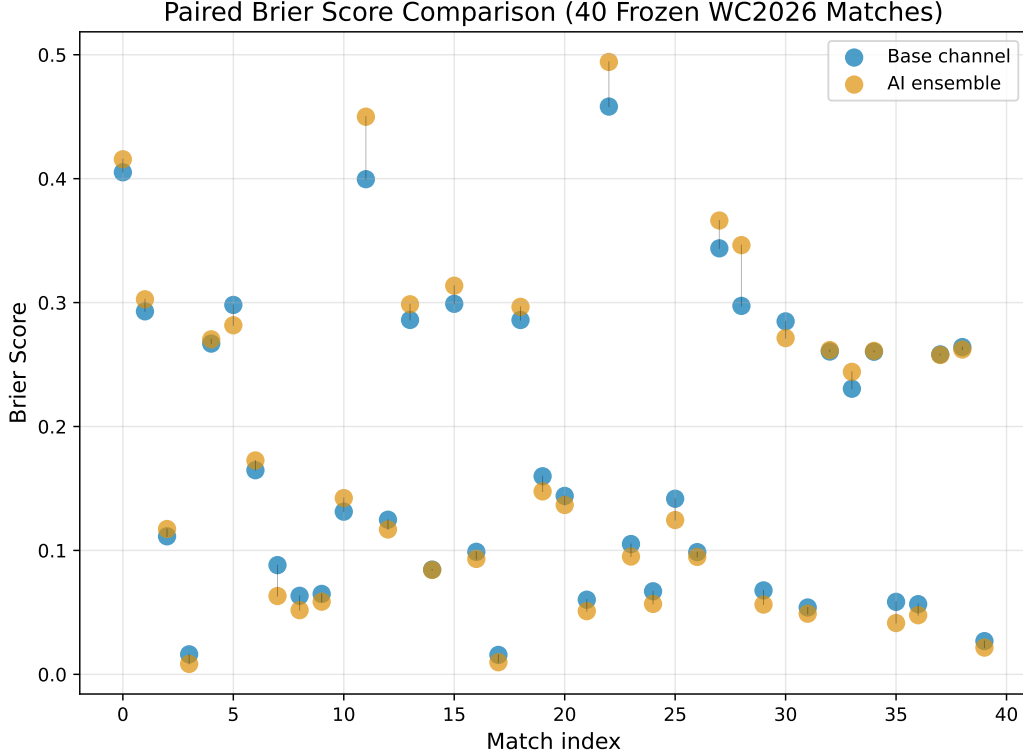


Figure 3: Per-match multiclass Brier scores for the base Elo-Poisson channel and the AI ensemble on the 40 frozen World Cup matches. Points below the diagonal favor the AI channel; points above favor the base channel.

5.3 Axis C: Model–market outright divergence

The outright market disagrees with the model most strongly at the probability extremes. Polymarket assigned Spain a 59.2% champion probability while the model assigned 26.0%; Argentina 41.3% versus 12.2%. Conversely, Polymarket compressed several strong European and South American teams to 0.05% while the model assigned 3–13%. Figure 4 shows the scatter; the Pearson correlation is $r = 0.828$ ($p < 0.001$) but the rank correlation is much weaker (Spearman $\rho = 0.285$, $p = 0.050$), driven by these extreme disagreements.

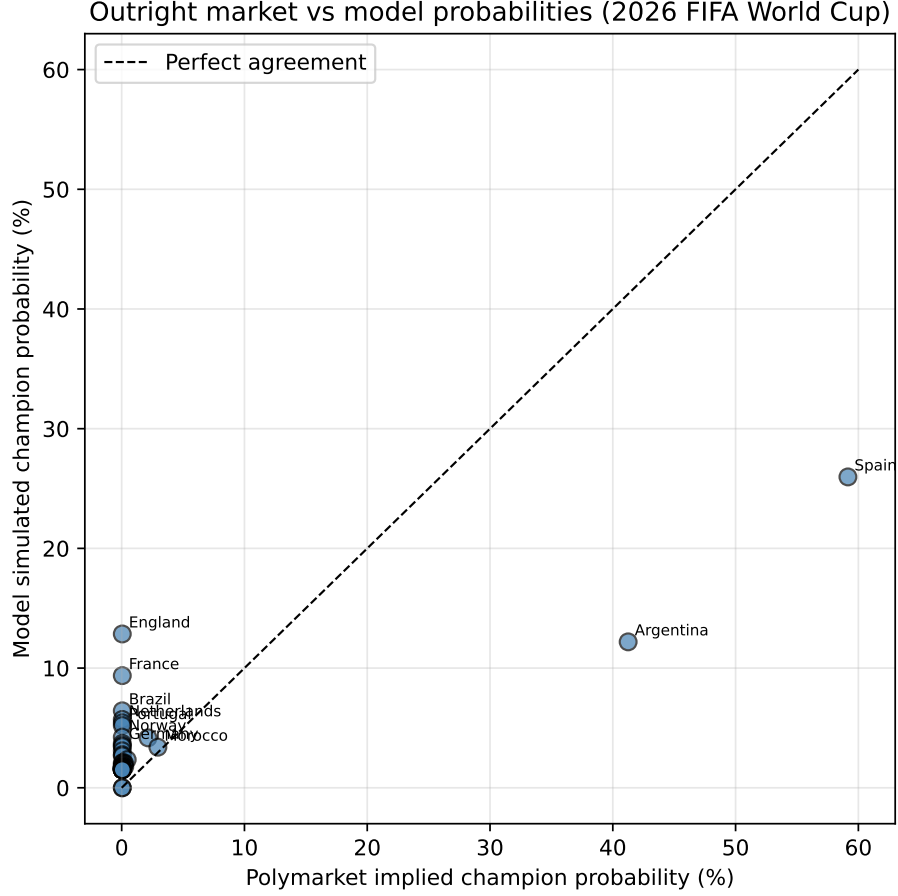


Figure 4: Model-simulated versus Polymarket-implied champion probabilities for all 48 teams. Labeled points are the ten teams that triggered divergence signals.

Across 22,390 divergence records, a pooled regression

$$\delta_{i,t} = \beta_0 + \beta_1 p_{i,t}^{\text{market}} + \beta_2 d_t + \beta_3 (p_{i,t}^{\text{market}} \times d_t) + \epsilon_{i,t}$$

where d_t is days to the final, yields $R^2 = 0.902$ and $\hat{\beta}_1 = -0.639$ ($p < 0.001$). Thus, as market-implied probability rises, the model's relative assessment of the team falls—the market is much more concentrated on favorites than the model. The time variables are weaker and harder to interpret because snapshots continue after the tournament ends. Figure 5 shows the divergence trajectory for the five teams with the largest mean absolute divergence.

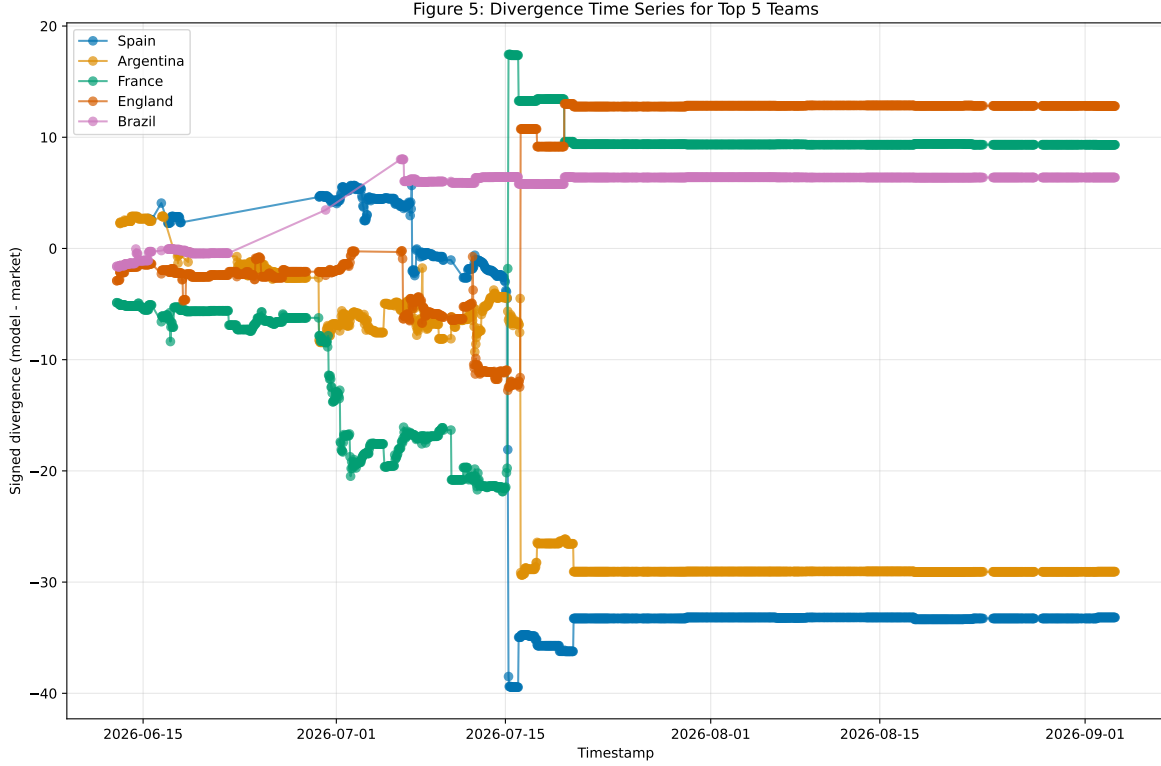


Figure 5: Signed divergence (model – market, percentage points) over time for the five teams with the largest mean absolute divergence. Snapshots after the final (2026-07-19) are post-tournament records and are included for completeness.

5.4 Axis D: Follow-through of divergence signals

We issued ten high-divergence signals before the tournament. Table 1 reports the group-stage record of each signaled team. Most signaled favorites performed well (Spain, Argentina, France, Brazil, England, Netherlands, and Morocco all advanced), while Germany and Norway did not. The sample is far too small to test statistical significance; the exercise is descriptive and is best read as a plausibility check rather than a trading strategy.

Table 1: Group-stage record of the ten teams flagged by the divergence engine.

Team	Wins	Draws	Losses	Signal direction
Spain	2	1	0	market over model
Argentina	3	0	0	market over model
England	2	1	0	model over market
France	3	0	0	model over market
Brazil	2	1	0	model over market
Portugal	1	2	0	model over market
Germany	2	0	1	model over market
Netherlands	2	1	0	model over market
Morocco	2	1	0	market over model
Norway	2	0	1	model over market

6 Discussion

The WorldCup Predict is a well-calibrated historical forecaster but not a dominant tournament predictor. Its 59.9% accuracy on 5,908 matches is a strong result, yet on the 40 frozen World Cup matches it reaches only 57.5% and the AI ensemble adds no measurable value. This pattern is consistent with the well-known difficulty of predicting single tournaments: historical averages are informative but do not capture team-specific form, injuries, or tactical matchups.

The market divergence results are more surprising. Polymarket’s outright prices were extremely concentrated on Spain and Argentina, compressing the implied probabilities of other strong teams to near-zero. Whether this reflects genuine information (e.g., inside knowledge, liquidity constraints, or contract structure) or a behavioral compression of probabilities is unclear from the data. The engine’s structural model assigns meaningful probability to a broader set of contenders. On a single tournament this disagreement cannot be resolved statistically; only repeated application across many tournaments could determine whether the model or the market is closer to true probabilities.

A secondary lesson concerns reproducibility. By freezing predictions and recording append-only snapshots, we transform a retrospective forecasting exercise into a preregistered natural experiment. This discipline is cheap to implement and sharply limits the risk of hindsight bias or overfitting to realized results.

7 Limitations

Single tournament and small frozen sample. The 40 frozen matches and 10 signals are a single realization. Effect sizes are noisy and confidence intervals are wide.

Partial knockout coverage. We have actual results for only 10 of the 32 knockout matches, so late-stage performance is unobserved.

Outright-only market comparison. Per-match odds from Polymarket or traditional book-makers were not continuously available; our market comparison is restricted to tournament-long outright contracts.

Model limitations. The engine does not condition on injuries, lineups, or in-game events. Its Elo ratings are historical and update slowly.

Causal interpretation. A divergence between model and market is not by itself evidence of market inefficiency; it may reflect information the model lacks or constraints the market faces.

8 Conclusion

We introduced WorldCup Predict, an open-source pipeline for probabilistic football forecasting, and subjected it and the Polymarket outright market to a transparent, frozen-prediction audit over the 2026 FIFA World Cup. The engine is historically well calibrated and beats a naive baseline by 13.4 percentage points, but a second-stage AI ensemble does not improve its frozen tournament predictions. The market and the model diverge systematically, with Polymarket concentrating probability on a small set of favorites. All code and data are released so that others can extend the analysis to future tournaments.

Reproducibility Statement

The complete source code for WorldCup Predict, the frozen pre-tournament predictions, the 1,619 market snapshots, the backtest records, and the analysis scripts for this paper are available at <https://github.com/tersemind/worldcup-predict>. The code is released under the MIT License, Copyright (c) 2026 TerseMind. No proprietary data or non-public API credentials are required to reproduce the figures and tables; all market data were obtained from public Polymarket and Kalshi endpoints.

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